Harsh Rana

+91 8452062333 | Mumbai, India
harshtrana0210@gmail.com | www.linkedin.com/in/harshrana02

EDUCATION

B.Tech - Artificial Intelligence and Data Science (CGPI - **9.46**) May 2020 - June 2024
Thakur College of Engineering and Technology, University of Mumbai *Mumbai, India*

EXPERIENCE

PwC India (Entity: **PWCA LLP)** Dec 2025 - Present
Associate (Artificial Intelligence Engineer) *Mumbai*

- Architected an event-driven, pan-India AI pipeline using **Python**, **Redis**, and **Azure Cosmos DB** that leverages **NLP** to parse transcripts and automate task distribution, containerized via **Docker** and deployed across **Azure Container Apps** and **Azure Container Jobs** utilizing **KEDA** for **event-driven autoscaling** with a React frontend.
- Engineered a high-throughput data pipeline using **Python**, **Redis**, and **Azure Cosmos DB** to automatically classify complaints against **government compliance**, deploying microservices via **Azure Container Apps** and orchestration tools like **KEDA** to enforce immutable **audit trails**.

Tata Consultancy Services Dec 2024 - Nov 2025
Systems Engineer (Artificial Intelligence Engineer) *Mumbai*

- Led **Generative AI** adoption across the **SDLC** by integrating **LLM** capabilities to drive **automation**, **rapid prototyping**, and **enterprise innovation**.
- Designed a **low-latency RAG pipeline** on **GCP** using **LangChain** and **FAISS** (migrated to **Vertex AI Vector Search**), serving insights via **Streamlit** while reducing query latency by **50%**.
- Designed a secure, **VNet-enclosed AI coding assistant** using **Azure OpenAI Service** and private **GPT-4.1** endpoints. Deployed via **Azure Private Link** to ensure offline, **enterprise-grade** developer support.
- Built an Azure-native **multimodal vision pipeline** (**Azure AI Vision**, **Azure AI Search**) for **zero-shot** cataloging.

Dytel Technology Group May 2023 - June 2024
Intern (Data Engineer) *Mumbai*

- Engineered a high-performance data pipeline with **Airbyte**, **Talend** and **PostgreSQL** that cut processing latencies by **30%** to support distributed **machine learning** and **predictive analytics** via **PySpark**.
- Delivered these model insights by embedding interactive **Metabase** dashboards directly into the client application, creating a seamless **user experience**.

PROJECTS

TraceLog Agent: Autonomous Log-Debugging Companion (Python, LangGraph, CrewAI, VS Code API, LLMs)

- Engineered a **resident AI developer agent** integrated natively into **VS Code** to perform **real-time monitoring** and **semantic analysis** of continuous terminal execution logs.
- Orchestrated a multi-agent architecture combining **LangGraph** and **CrewAI** to automatically isolate runtime anomalies, trace error stacks, and generate context-aware code remediations, drastically **reducing manual debugging time.**

Scholarship Chatbot (Python, OpenAI, LangChain, ChromaDB, CUDA)

- Built a **RAG**-powered scholarship chatbot by orchestrating **LangChain**, **Mistral LLM**, and **ChromaDB** with a **semantic retrieval pipeline** utilizing scraped JSON and **CUDA**-accelerated **OpenAI embeddings**.

SKILLS

Programming Languages	Python, SQL, Java
Cloud / Data Tools	AWS, Azure, GCP, PostgreSQL, MS SQL, Vector DB, Redis, Kafka, Kubernetes, Linux
Web Frameworks	FastAPI, Flask, REST APIs, Flutter, HTML/CSS, JavaScript
Generative AI & LLMs	LangChain, LangGraph, CrewAI, RAG, Ollama, llama.cpp

PUBLICATIONS

- Healthcare Data Analytics and Visualization Tools and Application [Taylor & Francis journal – **Scopus Indexed**]